

A referential montage is only as good as its reference electrode.

The reference electrode is intended to be at a distant and neutral compared to the active electrode which lies close to the potential focus (maximal potential) of cerebral activity. However, if the potential focus lies at the reference electrode or if the reference is contaminated by artifactual potentials, then it becomes an active reference. Such a reference is no longer neutral and may mislead the reader into false localization.

There are two distinct possibilities the reader should consider when suspecting an active reference:

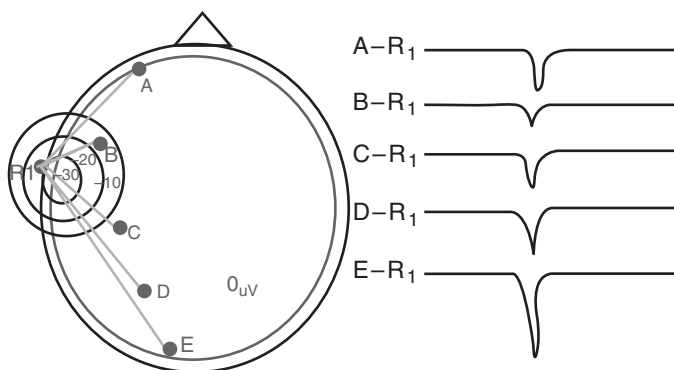
1. The reference lies at or near the potential focus.
2. The reference lies away from the potential focus but is contaminated by artifact.

When the reference is active because it is at or near the potential focus, all the channels in that referential montage will show deflections in the same direction. These may be of similar or different amplitudes but their deflections will all be in the same direction (waveforms will have the same polarities). In fact, those channels with the least amplitude lie closest to the focus (Figure 5.1).

The reference may also be active due to contamination by artifactual potentials. If a significant potential focus lies distant to the contaminated reference, an apparent phase reversal may occur at the channel closest to the focus. This is because both the contaminated reference electrode and the electrode close to the focus are active. As always, the direction of deflection at this channel will be governed by the law of polarity, that is, the pointer deflects toward the relatively negative electrode (Figure 5.2).

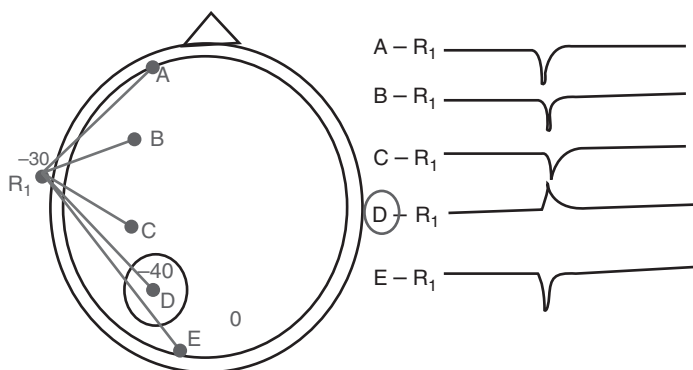
Each reference has its own common culprit artifact and no single reference is ideal for every situation. The choice of reference should be individualized based on the suspected location of the waveform and the contaminants common to that point of reference.

Central reference (Cz) is good for evaluating temporal waveforms but commonly contaminated by sleep potentials. Hence, Cz is best employed to evaluate temporal patterns during wakefulness. Ear references (such as



Potentials in microvolts (μV)

Figure 5.1 An active reference due to a hypothetical focus near the reference (R_1); note the positive (downward) deflection in all channels. Channel B- R_1 has the least deflection, as electrode B lies closest to the focus, while channel E- R_1 has the largest deflection, as electrode E is most distant from the focus.



Potentials in microvolts (μV)

Figure 5.2 An active reference due to an artifactual potential ($-30 \mu\text{V}$) at R_1 . Additionally, there is a distant potential ($-40 \mu\text{V}$) at electrode D. Note the negative (upward) deflection in channel D- R_1 , as D is relatively more negative than R_1 . There are positive deflections (downward) in the other channels due to the negative contaminant potential at R_1 . This gives the appearance of a phase reversal on this ipsilateral ear referential montage.

ipsilateral and contralateral ear) are good for evaluation of central waveforms but they are commonly contaminated by EKG and temporalis muscle artifacts. Temporal discharges may also originate close to the ear reference. Therefore, ear references are best employed to evaluate central patterns during drowsiness or sleep. Similarly, a frontal reference (Fz) is good to evaluate an occipital pattern but commonly contaminated by eye movement artifact while occipital references (Oz) are good for frontal patterns but are contaminated by the posterior-dominant rhythm. If an active reference is detected, then an alternative referential montage should be used to confirm localization. Digital displays make switching between differential montages easy [1,2].

Chapter Summary

1. Referential montages are only as good as their reference electrode.
2. The reference electrode is intended to be distant from the potential focus and inactive, but sometimes it may be very active. This is called an active reference; it misleads the reader into false localization.
3. An active reference occurs if the reference electrode lies close to the potential focus or becomes contaminated by artifactual potentials.
4. If the reference is at or near the focus, typically there is a unidirectional reflection (often with varying amplitudes) in all the channels.
5. If the reference is contaminated by an artifactual potential, typically there is a phase reversal over channels that overlie a potential focus.
6. Central references are prone to contamination with sleep potentials, and ear references are prone to contamination with EKG and temporalis artifact.
7. Switch to an alternative referential montage if you suspect an active reference. This is easy with a digital display.

References

1. Teplan M. Fundamentals of EEG measurement. *Measurement Science Review*. 2002 Jan;2(2):1.
2. Lagerlund TD. Manipulating the magic of digital EEG: montage reformatting and filtering. *American Journal of Electroneurodiagnostic Technology*. 2000 Jun 1;40(2):121–36.